

ENGR 3323: Signals and Systems

Sample of Test 1

Time (80 minutes): 11:00 to 12:20

Name:

Note: Show all steps of your solution for full credit. A final answer is not enough.

Q1) Evaluate the integral $\int_{-\infty}^{\infty} t^2 \delta(t - 2) dt$

Q2) Plot $x(2t-3)$ where $x(t)$ is shown in **Fig. 1** below.

$$x(t) = 2t \quad 0 \leq t \leq 2$$

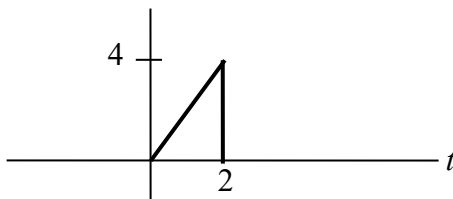


Fig. 1

Q3) Find and plot the derivative, dx/dt , of the signal $x(t)$ shown in **Fig. 1** above.

Q4) A system described by $y(t) = t x(t - 1)$. Is the system causal?

Q5) The impulse response of a realizable system is $h(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$, where $C_1 = 2 - j3$ and $\lambda_1 = -2 + j10$.

- a) Find C_2 and λ_2
- b) Express the impulse response in this format $h(t) = A e^{\sigma t} \cos(\omega t + \theta)$. Make sure to find the numerical values of the parameters A , σ , ω , and θ .

Q6) Model the system shown below

Q7) Find the zero-input response of LTIC system described by the differential equation below with initial conditions $y(0) = 2$ and $dy(0)/dt = 1$:

$$(D^2 + 6D + 8)y(t) = (D^2 - 2)x(t)$$

Q8) Find the unit impulse response, $h(t)$, for an LTIC system described by the following differential equations:

$$(D^2 + 8D + 16)y(t) = (D - 1)x(t)$$